Investment Appraisal

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An introduction to investment appraisal

- Investment appraisal involves comparing **the expected future cash flows of an investment with the initial outlay** for that investment
- A business may want to analyse:
  - **How soon** the investment will recoup the initial outlay
  - **How profitable** the investment will be
- Before an investment can be appraised, key data will need to be collected, including:
  - Sales forecasts
  - Fixed and variable costs data
  - Pricing information
  - Borrowing costs
- The collection and analysis of this data is likely to **take some time**
  - It requires **significant experience** to interpret the data appropriately **before the investment appraisal can take place**
- Different methods are **used to appraise the value of an investment**, including:
  - The simple payback period
  - The average rate of return (ARR)
  - The net present value of the discounted cash flow

Simple payback period

- The payback period is a calculation of the **amount of time** it is expected an investment will **take to pay for itself**
- Where net cash flows are expected to be **constant over time**, the payback period can be calculated using the formula

$$\text{Payback period} = \frac{\text{Initial outlay}}{\text{Net cash flow per period}} = \text{Years / Months}$$

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Got it

facility at a cost of £200,000. It expects an additional net cash flow of £30,000 per year as a result of the investment.

Calculate the payback period for the investment.

[3]

Step 1: Divide the initial outlay by the additional expected net cash flow

$$= \frac{£200{,}000}{£30{,}000} \quad [1]$$

$$= 6.67 \text{ years}$$

Step 2: Convert the outcome to years and months

- 6 years
- 0.67 years = 8.04 months [1]
- **Payback period = 6 years and 8 months [1]**

Worked Example

Help me understand

Hammer and Son provides a household repairs service that has recently employed a new handywoman who requires her own van. The new van will be purchased for £32,000.

The net cash flows are expected to vary over the five years following its purchase and are shown in the table below.

Year	Net cash flow (£)	Cumulative cash flow (£)
0	(32,000)	(32,000)
1	14,000	(18,000)
2	10,000	(8,000)
3	6,000	(2,000)
4	3,000	1,000
5	2,000	3,000

Calculate the payback period for the van.

[4]

Step 1: Identify the final year where the cumulative cash flow is negative

- In this case, the cumulative cash flow figure is -£2,000 at the end of Year 3
- This is the remaining amount (outlay) outstanding [1]

Step 2: Calculate the monthly net cash flow for the next year

$$= £3{,}000 \div 12 \text{ (months)} \quad [1]$$

$$= £250$$

$$= £2{,}000 \div £250$$ [1]

$$= 8 \text{ months}$$

Step 4: Identify the payback period

- In this case, the payback period is 3 years and 8 months [1]

Evaluation of the payback period method

Benefits	Drawbacks
• It is a **simple method** to calculate and understand • It is particularly useful for businesses where **cash flow management** is vital • Businesses can **identify the point** at which an investment is paid back and is **contributing positively** to cash flow • It is also useful when **new technology** is introduced regularly • Businesses purchasing equipment can calculate whether an investment "pays back" **before an upgrade is available**	• It provides **no insight** into the **profitability of investments** • Payback only considers the total length of time to **recover an investment** • Neither the timing nor the **future value of cash inflows** is considered • It may encourage a **short-termism** approach • Potentially lucrative **investments may be dismissed**, as they take longer to pay back than alternatives

Average rate of return

- The average rate of return (ARR) compares the average **profit** per year generated by an investment with the value of the **initial outlay**
- The average rate of return is calculated using the formula

$$\text{ARR} = \frac{\text{Average annual return}}{\text{Initial outlay}} \times 100$$

- The outcome of the formula is **expressed as a percentage**, which makes it easy to **compare different investment options**

Worked Example

Creative Frames, a small artwork framing business, is considering an investment of £40,000 in new machinery.

Calculate the average rate of return of the proposed investment. [4]

Step 1: Calculate the total profit over the lifetime of the investment

$$\text{Total cash inflows} - \text{Total cash outflows} = \text{Total profit}$$ [1]

$$= £140{,}000 - £92{,}000 = £48{,}000$$

Step 2: Divide the total profit by the number of years of the investment project to find the average annual profit

$$= £48{,}000 \div 6 \text{ years}$$ [1]

$$= £8{,}000$$

Step 3: Divide the average annual profit by the initial outlay

$$= £8{,}000 \div £40{,}000$$ [1]

$$= 0.2$$

Step 4: Multiply the outcome by 100 to find the percentage

$$= 0.2 \times 100$$ [1]

$$= 20\%$$

Evaluation of the average rate of return

Benefits	Drawbacks
• It considers **all of the net cash flows** generated by an investment over time • It is easy to understand and compare the percentage returns with each other	• As it depends on an average of cash flows, **it ignores the timing of those cash flows** • The **opportunity cost** of the investment is ignored, as values are neither expressed in **real terms** nor are adjustments made for the impact of interest rates and time

Net present value of discounted cash flow

- Net present value (NPV) is a financial metric used to evaluate the value of an investment or a project
- The NPV of an investment takes into account the **effects of interest rates and time**
  - It represents the **present value** of the **future cash inflows** minus the **present value** of the **future cash**

- This discounting method takes into account:
  - The fact that money received in the future is worth less than money received today due to inflation
  - The **opportunity cost** of not having the money available for other uses

- To calculate the NPV of an investment, **the value of all future net cash flows in today's terms** needs to be calculated first — and then **discounted using a table**
  - The **cost of the initial investment is deducted** from the total of the discounted net cash flows:
    - If the sum of future net cash flows minus the initial investment is **positive**, then the investment is **likely to be worthwhile**
    - If the sum of future net cash flows minus the initial investment is **negative**, then the investment is **unlikely to be worthwhile**

- Discounted cash flows are calculated using **discount tables**, which allow future cash flows to be **expressed in today's terms**

Worked Example

Brownsea Sightseeing Tours Ltd is considering purchasing a new pleasure craft at a cost of £325,000.

It expects the investment to achieve the following net cash flows over five years of operation:

Year	Net cash flow (£)	10% discount factor
0	(325,000)	1.00
1	110,000	0.91
2	90,000	0.83
3	75,000	0.75
4	65,000	0.68
5	60,000	0.62

Using the 10% discount factor, calculate the NPV of the leisure craft investment.

Step 1: Calculate the discounted cash flow for each year by multiplying the net cash flow by the discount factor

Step 2: Total the discounted cash flow values for each year, including Year 0

$= (£325{,}000) + £100{,}100 + £74{,}400\ £56{,}250 + £44{,}200 + £3$

$= (12{,}550)$

- The net present value of the investment is -£12,550
- Given that it is negative, this suggests that the investment in the new pleasure craft is **not financially worthwhile**

Evaluating the net present value method

Benefits	Drawbacks
• It considers the **opportunity cost** of money • Discount tables are used to **calculate the forecasted future values** of net cash flows • Businesses may choose **different discount tables (20%, 10%, 5%, etc.)** to adjust the level of **risk involved** in a project, allowing a **range of scenarios** to be considered	• It is more **complicated** to calculate and interpret than other methods of investment appraisal • One of the primary challenges of using the NPV method is **accurately forecasting** future cash flows • Selecting an **appropriate discount rate** can be challenging, and even small changes in the discount rate can significantly impact the calculated NPV • The NPV method only **considers the financial costs** and benefits of a project and does not account for non-financial benefits or costs, e.g. environmental damage

Examiner Tips and Tricks

To account for inflation, the discount rate used in the NPV calculation should be adjusted to reflect the expected inflation rate.

Limitations of investment appraisal techniques

- Each of the investment appraisal techniques **relies upon forecasted future cash flows**, which may **lack accuracy**:
  - Managers compiling cash flow forecasts may lack experience or may be biased towards a particular investment
  - Incomplete past data may make forecasting imprecise or mean that confidence in the data used to compile the forecast is limited
- **Long-term cash flow forecasts** can be **inaccurate** for several reasons:
  - Unexpected increases in **costs**
  - The arrival of **new competitors**
  - Changes in **consumer tastes**
  - Uncertainties due to economic growth or **recession**
- Factors other than the cost of investment and the return on investment are not considered:
  - Business finances and the availability of external finance to fund the investment
  - The overall **corporate objectives**
  - Potential for positive **public relations** or meeting **social responsibilities**

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